Counterfactual Concept Swaps on InceptionV1 Curve Detectors: a Specific but Partial Causal Mediator of Curve Orientation

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Abstract

Circuits-style neuron explanations in vision models rest mostly on observational evidence. We test one of the best-characterised, "the ten InceptionV1 `mixed3b` units are curve detectors tuned to orientation", with pre-registered counterfactual interchange interventions (concept swaps). After reproducing the authors' published tuning curves ($r = 0.969$), we copy the family's activations between images that differ only in curve orientation and read the next layer's published curve units. The family is a highly *specific* mediator: it moves the next layer's curve code 0.410 of the way to the source, versus a 95th percentile of 0.098 for 2000 activation-change-matched 10-channel controls. It carries curvature, flipping a line to "arc" in 0.807 of pairs, and its ablation changes the network's predictions on natural images about twice as often as matched ablations (0.172 vs 0.093). It is *not sufficient*: the next layer decodes the source orientation in only 0.177 of pairs [0.156, 0.199], so the pre-registered strong hypothesis is falsified. The orientation code is distributed and redundant (complement interchange accuracy 0.403; about 80 channels for reliable transfer), and effects add almost linearly. The same specific-but-partial pattern holds for the 3a→3b curve circuit, on natural backgrounds, and under a natural rotation counterfactual. Post-hoc checks leave these conclusions unchanged: a stricter rank-matched null, the decoded-orientation metric, and blur-free rotations.

1 Introduction

Neuron-level explanations of vision models are mostly *observational*. A unit is called a "curve detector" because it fires on curves, because its feature visualization looks like a curve, and because its weights connect to other curve-like units [6, 7, 25]. In language models, Huang et al. [19] showed that even confident neuron explanations can have high observational error rates and little to no causal effect when tested *interventionally*. In their test, the explanation is a causal claim that the unit mediates the concept: copying the unit's value from a source input into a base input should make the model behave as if it saw the source. Related "concept swap" tests now validate representations in language models: digit-position neurons [2], SAE and attribution-graph features [1, 5], and the Jacobian-lens workspace [17].

We apply this interventional standard to the best-characterised neuron family in vision: the ten `mixed3b` curve detectors of InceptionV1 [6, 15]. The explanation names a concrete variable, curve orientation. That makes exact counterfactual pairs possible: two images identical except for orientation. It also lets us check our setup against the authors' published tuning data before any intervention.

Contributions.

- We reproduce the published tuning curves of the 3a, 3b and 4a curve families. The fit is $r = 0.969$ for 3b and $r = 0.847$ for 4a, with the orientation convention fitted on 3a only (§4.1).
- We run pre-registered interchange interventions with activation-change-matched and branch-matched null neuron sets. Readouts are the published `mixed4a` curve family and, separately, all of `mixed4a`. We also test natural images.
- The main result is mixed. The curve family moves the next layer's orientation code 0.410 of the way to the source; 95% of activation-change-matched 10-channel controls move it less than 0.098, and none transfers the decoded orientation in more than 0.13% of pairs. Yet swapping the family makes the next layer decode the source orientation in only 0.177 of pairs. By our pre-registered rule, the strong sufficiency claim is falsified. The family is a highly specific but partial mediator, and other `mixed3b` channels redundantly carry the orientation.

2 Background and related work

Circuits and curve detectors. Olah et al. [25] introduced the circuits framing, and Olah et al. [26] catalogued InceptionV1's early-vision families. Cammarata et al. [6] characterised ten `mixed3b` curve units with synthetic curves and radial tuning curves. Cammarata et al. [7] reverse-engineered how 3a curves build 3b curves. Olah et al. [27] described the family as rotated copies of one feature, and Voss et al. [31] located curve units in the 5×5 branch. Gorton [15] found additional curve features with sparse autoencoders. Cammarata [8] revisits layer 3b and argues that neurons are cleaner than superposition accounts suggest. None of these works runs counterfactual activation swaps on the family.

Evaluating neuron explanations. Network dissection labels units and ablates them [3, 4]. Class selectivity does not predict ablation importance [21, 23], and feature visualizations can mislead [13]. Synthetic-image checks [20] and metric sanity checks [24] evaluate explanations observationally. Recent work frames unit interpretation as falsifiable hypothesis testing [10].

Interchange interventions. Causal abstraction [11, 12] formalises when a set of units realises a high-level variable. Its operational test is the interchange intervention, also called activation patching. We follow the metric advice of Zhang and Nanda [32] and Heimersheim and Nanda [18]: a binary success measure (IIA) plus a continuous, roughly linear effect. Resampling-based hypothesis tests [9] motivate our complement and null sets.

3 Method

Model. We use the Lucid InceptionV1 ("inception5h"; 30), loaded with Lucent. Its local response normalisation is set to the TensorFlow-exact constants (window 11, $\alpha = 11 \cdot 10^{-4}$, $\beta = 0.5$, $k = 2$; Lucent PR #51). All convolutions run in fp32 with TF32 disabled.

Stimuli and splits. Arcs are rendered with 4× supersampling. Each spans 120°, and its midpoint sits on the receptive-field (RF) centre of the readout units, pixel (111,111), found from the input-gradient centre of mass [22]. The orientation θ is the direction from the midpoint to the centre of curvature. A split is 72 orientations × radii $\{30, 45, 60, 80\}$ × widths $\{3, 5\}$ × two polarities, giving 1152 images. Each nuisance cell gets one random translation of ± 2 px, and the jitter seed defines

the split: selection (seed 1) and confirmatory (seed 2). Each base gets four sources from its own cell with $|\Delta\theta| \geq 30°$, giving 4608 minimal pairs.

Intervention. The readout is `mixed4a` unit (6,6). It depends only on an 11×11 window of `mixed3b`: the 5×5 convolution sees `maxpool4` cells 4–8, which pool `mixed3b` rows 8–18. For a neuron set S we overwrite channels S of the base window with the source's values and recompute `maxpool4`→`mixed4a` on the window. This equals the full network to 10^{-14} in fp64 (unit test).

Readouts τ and metrics. The primary readout uses the 14 published `mixed4a` curve units (pre-ReLU, z-scored on the selection split). Orientation is decoded by maximum Pearson correlation with the 72 clean selection templates of the pair's nuisance cell. The interchange intervention accuracy is

$$\mathrm{IIA}(S) = \frac{1}{|P|} \sum_{(b,s)\in P} \mathbf{1}\Big[\big|\tau(M_{S\leftarrow s}(b)) - \tau(M(s))\big|_\circ \leq 10°\Big], \tag{1}$$

taken over eligible pairs P: those whose clean base and clean source both decode within 10° of their true orientation. The continuous *normalized effect* is $\sum\langle z_p - z_b, z_s - z_b\rangle / \sum\|z_s - z_b\|^2$, over pairs outside the lowest 5% of $\|z_s - z_b\|^2$. It is 0 for no change and 1 when the readout becomes the source's. The secondary, non-circular readout is a ridge regression from all 508 `mixed4a` channels to $(\cos\theta, \sin\theta)$.

Controls. A patch acts only through channels whose values differ between base and source, so a random null is too easy to beat. Our primary null draws 2000 non-family sets matched to the family's histogram over deciles of Δ_c, the mean squared base→source change of channel c on selection pairs. A co-primary null draws 2000 sets from the rest of the 5×5 branch. References are the Fisher-score top-10 channels for θ [2, 16], the complement (470 channels) and all 480 channels.

Statistics. Pairs share base and source images, so we use the pigeonhole bootstrap [28] ($B = 2000$), recomputing the null's 95th percentile in every replicate. p-values are $(1+\#\{\text{null} \geq \text{obs}\})/(1+M)$ [29], and secondary tests are Holm-corrected.

Pre-registered decision rule. Written before the confirmatory run, with pilot results disclosed. Let $D = \mathrm{IIA}_{\text{family}} - q_{95}(\text{null})$. H1 is *supported* if IIA ≥ 0.5, $D \geq 0.10$, and the 95% CI lower bound of D is above 0 against both nulls. It is *falsified* if the CI upper bound of D is below 0.05 or IIA < 0.2; otherwise it is inconclusive. Each secondary hypothesis is supported if its normalized-effect margin over the Δ-matched q_{95} has a CI lower bound above 0 and a Holm-adjusted $p < 0.05$.

4 Results

4.1 E1: the observational explanation replicates

We fitted one orientation convention (a reflection plus a 180° shift) on `mixed3a` alone, with mean $r = 0.937$. Under that convention, our radial tuning curves match the published ones with mean Pearson $r = 0.969$ for the 10 `mixed3b` units (every unit between 0.96 and 0.98) and $r = 0.847$ for the 14 `mixed4a` units. The 95th percentiles of a unit-label permutation null are 0.254 and 0.242, and $p = 10^{-4}$ for each (Fig. 1). Each 3b unit separates arcs within ±45° of its preferred orientation from other orientations with AUROC 0.70–0.95, and from straight lines with AUROC 0.67–0.92. With Lucent's default LRN constants instead of the TF-exact ones, the correlations are essentially

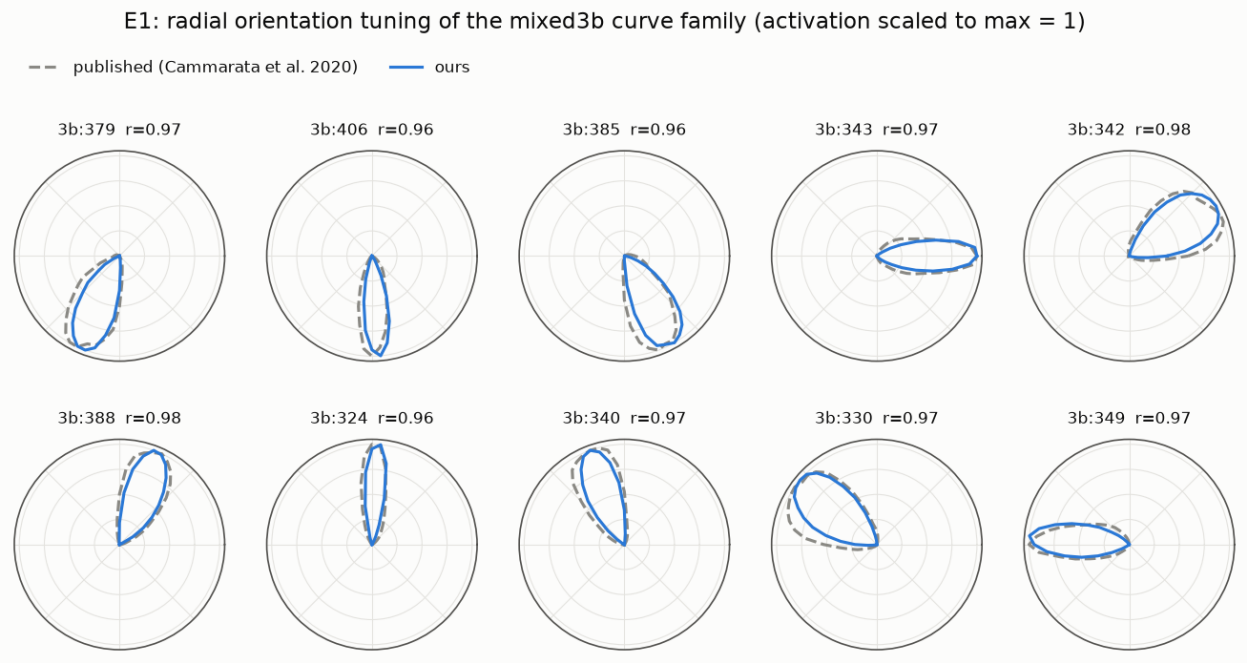


Figure 1: E1. Radial orientation tuning of the 10 `mixed3b` curve units: ours (blue) and published by Cammarata et al. [6] (gray, dashed), each scaled to a maximum of 1. The orientation convention was fitted on `mixed3a` only.

unchanged (0.969 for 3b, 0.848 for 4a). Our model port, renderer and readout therefore reproduce the evidence the explanation rests on.

4.2 E2: the family is specific but not sufficient

H1 (primary). Clean decoding accuracy is 0.996, leaving 4574 of 4608 pairs eligible. Swapping the 10 curve units yields IIA = 0.177 [0.156, 0.199]. This is below the pre-registered falsification bound of 0.2, so **H1 is falsified**. The verdict is identical for two further jitter seeds (IIA = 0.172, 0.171) and with Lucent's default LRN (0.172).

Specificity. The same swap is far more potent than any matched control (Fig. 2). The Δ-matched null has IIA $q_{95} = 0.001$ and the branch-matched null $q_{95} = 0.000$; across 4000 sets the best transfers orientation in 0.13% of pairs ($p = 0.0005$ against each null). The normalized effect of the family is 0.410 [0.400, 0.420], against a Δ-matched q_{95} of 0.098 and a branch-matched q_{95} of 0.077. The data-driven Fisher top-10, which shares 4 units with the family, reaches only IIA = 0.028. Mean-ablating the family leaves the base orientation decodable in 0.806 of pairs, against 0.985 for Δ-matched ablations. A same-orientation sham leaves it decodable in 0.999.

Redundancy and additivity. The 470 complementary channels reach IIA = 0.403 and effect 0.590. The two effects sum to 1.000. Over random K-subsets of the family, the effect grows almost linearly: $0.040, 0.080, 0.119, 0.202, 0.285, 0.410$ for $K = 1, 2, 3, 5, 7, 10$. IIA, in contrast, stays near 0 until $K = 7$ (0.045) and reaches 0.177 at $K = 10$. Additivity is partly structural: 6 of the 14 readout units belong to the 1×1 branch, whose pre-ReLU output is linear in the max-pooled `mixed3b` channels. Patching the top-K channels by activation change gives IIA $= 0.359, 0.442, 0.792, 0.979$

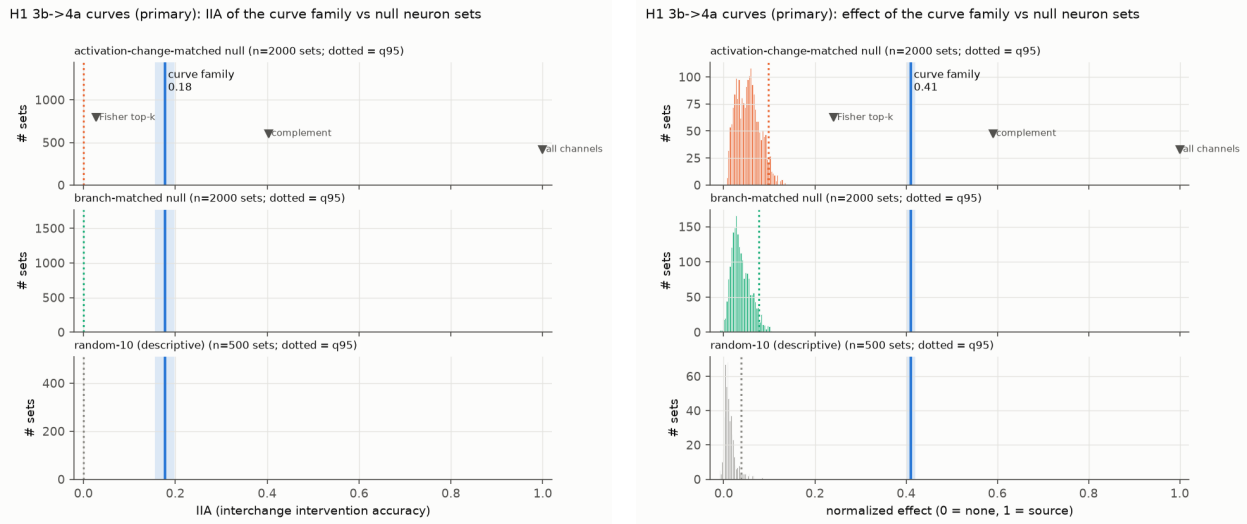


Figure 2: H1. IIA (left) and normalized effect (right) of the curve family (blue line, 95% CI band) against 2000 Δ-matched, 2000 branch-matched and 500 random 10-channel null sets. Triangles mark the reference sets.

for $K = 20, 40, 80, 160$. Reliable transfer therefore needs about 80 channels, and the 10 curve units are all among the top 17.

Secondary tests. All five secondary hypotheses meet the pre-registered criterion. The Holm-adjusted p-values are ≤ 0.004 (Table 1; Fig. 3).

- **Ridge readout over all 508 `mixed4a` channels.** This is non-circular. As pre-registered, the effect is the projection over all 508 z-scored channels: 0.248 against a null q_{95} of 0.106, with IIA = 0.105. It therefore measures whole-layer movement. Post hoc, we also measure it on the decoded orientation (§4.4).
- **Curvature swap (H2).** The family's effect is 0.393 against 0.141. Inserting the family's arc activations into a line image makes the 4a curve units read "arc" in 0.807 of pairs. The reverse direction succeeds in only 0.003 of pairs: other channels also signal curvature.
- **The Curve Circuits claim, 3a→3b (H4).** The effect is 0.324 against 0.169. IIA is only 0.058 while the complement reaches 0.757, the same specific-but-partial pattern as H1.

In the exploratory analyses, IIA depends on the orientation difference: 0.111 for $|\Delta\theta|$ in 30–60° and 0.303 in 120–150°. A J-lens/SAE-style *family rotation* swap gives each unit the map of its neighbour on the orientation circle. It shifts the decoded orientation in the predicted direction, but by only 21–39% of the predicted angle (e.g. $-8°$ vs $-36°$ for $k = 1$).

4.3 E3: natural images

With arcs composited onto Imagenette photos, the H1 pattern recurs: IIA = 0.190 and effect 0.415 against a null q_{95} of 0.104. In the natural rotation counterfactual (H5) on 506 curve-present photos, the patched base is read as the rotated version in IIA = 0.072 of pairs. Its effect is 0.287 against a null q_{95} of 0.075, while the complement dominates at IIA = 0.847. On curve-absent crops the family's effect halves to 0.135 (exploratory). End to end, mean-ablating the family over the whole

Table 1: Pre-registered tests. IIA and normalized effect of the curve family, with 95% pigeonhole CIs, against the activation-change-matched null's q_{95}. Verdicts: H1 rows use the pre-registered three-way rule; the other rows use the secondary criterion (effect margin CI > 0, Holm $p < 0.05$). Full table: `reports/generated/results.md`.

Test	IIA family	null q_{95}	Effect (null q_{95})	Verdict
H1: 3b→4a curve units	0.177 [0.156, 0.199]	0.001	0.410 (0.098)	falsified
H1-ridge: 3b→all 4a	0.105 [0.087, 0.123]	0.000	0.248 (0.106)	supported
H2: curvature swap	0.405 [0.371, 0.440]	0.054	0.393 (0.141)	supported
H4: 3a→3b curve units	0.058 [0.045, 0.071]	0.005	0.324 (0.169)	supported
Composited (photos)	0.190 [0.169, 0.211]	0.001	0.415 (0.104)	supported
H5: natural rotation	0.072 [0.054, 0.091]	0.000	0.287 (0.075)	supported
H1, jitter seed 3	0.172 [0.151, 0.193]	0.001	0.410 (0.094)	falsified
H1, jitter seed 4	0.171 [0.149, 0.193]	0.001	0.407 (0.104)	falsified
H1, Lucent LRN	0.172 [0.150, 0.194]	0.000	0.406 (0.096)	falsified

map changes the network's 1008-way top-1 prediction on 0.172 of 1962 photos. Variance-matched null ablations change it on 0.093 (q_{95} 0.110, $p = 0.0099$), and the mean KL is 0.2054 against 0.0501.

4.4 Post-hoc robustness (not pre-registered)

After the confirmatory run, an independent code review raised three ways the pre-registered comparisons could flatter the family. We tested each.

- **A softer-than-intended null.** All 10 curve units lie above the 96th percentile of activation change, so decile matching mostly draws channels that change less. A rank-matched null pairs each unit with one of its 10 nearest non-family channels by activation change (1000 sets). Against it, H1's effect 0.410 still far exceeds the null q_{95} of 0.121. The 10 non-family channels with the largest activation change reach only 0.125 and IIA 0.000. All other tests keep margin CIs above 0 (`reports/posthoc_robustness.md`).
- **Z-space vs decoded orientation.** Measured in the ridge decoder's $(\cos\theta, \sin\theta)$ output space, the family moves the decoded orientation vector 0.573 of the way to the source's (rank-null q_{95} 0.126). It lands within 10° in only 0.105 of pairs, so the decoded orientation rotates only part of the way.
- **Rotation blur in H5.** Rotating square crops by $-90/90/180°$ is an exact transpose with no interpolation. Under these rotations the H5 effect is 0.336 (null q_{95} 0.077) and IIA 0.108, both larger than with the original, partly blurred rotations.

5 Discussion

What the explanation gets right. The Circuits description of the `mixed3b` curve units survives every observational check we ran. Interventions add something observation cannot show: the units are causally *specific*. Among all 10-channel sets whose activations change as much between base and source, only the curve family moves the next layer's curve code substantially. It also carries curvature (H2), and removing it alters the network's decisions on natural images more than removing matched channels does.

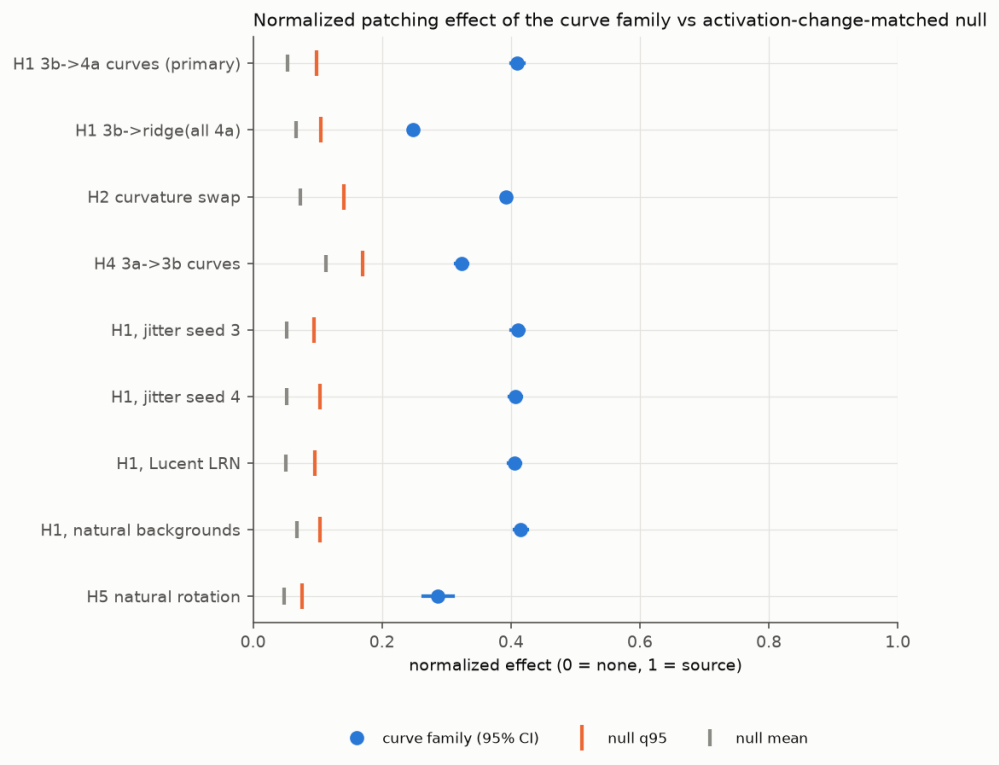


Figure 3: Normalized effect of the curve family (blue, 95% CI) for every test, against the Δ-matched null's mean (gray) and q_{95} (orange).

What it gets wrong, or leaves out. The explanation is often read as "these neurons are where the network represents curve orientation". Under Huang et al.'s criterion, that reading fails: the family is not a sufficient mediator. Orientation is carried redundantly by many `mixed3b` channels, about 80 for reliable transfer. The family's contribution adds roughly linearly to theirs. This agrees with Gorton [15], who found curve features outside the 10 neurons, and with the language-model finding that single neurons rarely mediate concepts alone [19]. In a causal-abstraction framing [12], the orientation variable is realised by a distributed subspace of which the family is a privileged but partial part.

Implications. Neuron-level explanations in vision should be stated with an explicit causal scope. Two claims need separating: "unit u is selective for X" (observational), and "unit set S mediates X for downstream computation Y" (interventional, with a stated readout). Activation-change-matched nulls are essential. Against random channels, almost any orientation-tuned set would look causally important, because most channels barely change between base and source.

6 Limitations

- **Single model and layer pair.** Only InceptionV1 and the 3b→4a (and 3a→3b) transition are tested. We do not claim generality to other CNNs or depths.
- **Partly circular readout.** The primary readout uses curve units the authors relate to 3b curves. We mitigate this with a whole-layer ridge readout and natural-image tests, but

both decoders are trained on clean data and judge patched, possibly off-manifold states. A same-orientation sham keeps the readout intact in 0.999 of pairs, which argues against patching artefacts, but it does not rule out decoder bias.

- **Strict binary metric.** IIA needs the decoded orientation within 10° of the source. Partial transfer appears only in the continuous effect. The pre-registered rule judges H1 on IIA, which makes it a statement about strong sufficiency.
- **Stimulus distribution.** The synthetic arcs are off-distribution for an ImageNet model. The composited and rotation experiments only partly address this. Rotating a photo changes far more than curve orientation, so H5's IIA is a conservative test.
- **No learned subspaces.** We test the published neuron set, not an optimised mediator such as distributed alignment search [14] or SAE features [15]. Whether a low-dimensional sufficient mediator exists is open.
- **Bootstrap assumptions.** The pigeonhole bootstrap treats base and source as independent factors, although every image plays both roles, and it ignores clustering within the 16 jitter cells. In H2, lines at θ and $\theta + 180°$ are near-duplicates. Our CIs are therefore somewhat too narrow. The three jitter seeds agree to within 0.006 IIA, and no verdict is close to its threshold.
- **Design details.** H5's clean decoding accuracy is 1.0 by construction, because its templates are the photo's own clean versions; success still requires the patched base to resemble the rotated version more than the unpatched base. The composited test reuses the same 32 backgrounds in both splits, so it generalises across jitter, not backgrounds. The 5% denominator filter for the effect was computed over all pairs rather than only eligible ones, as the pre-registration states. At most 1.7% of pairs are ineligible in any test (jitter seed 4), so the difference is negligible.
- **Pilot disclosure.** The pilot already suggested IIA ≈ 0.16. Thresholds were not changed, and the pilot is disclosed in the pre-registration.

7 Conclusion

Counterfactual concept swaps turn the Circuits curve-detector explanation into a falsifiable causal claim. The claim splits into two parts. The ten `mixed3b` curve units are a *specific* causal mediator of curve orientation and curvature: they far exceed activation-matched controls on synthetic and natural images and matter for the network's predictions. They are *not* a sufficient one: swapping them alone transfers the decoded orientation in only 0.177 of pairs, because the code is distributed and redundant. We release the pre-registration, code, per-pair data and figures so these numbers can be checked and extended.

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A Reproducibility and statistics details

Code, tests and the pre-registration are at https://github.com/BurnyCoder/counterfactually-causal-analysis-of-convolutional-neural-networks-with-concept-swaps. Per-pair data are in the Hugging Face dataset https://huggingface.co/datasets/BurnyCoder/inceptionv1-curve-concept-swaps. All numbers in this paper are LaTeX macros generated from the result files (reports/generated/numbers.tex).

Runtime is about one hour on an RTX 5070 Laptop GPU. Determinism: TF32 is disabled, and stimuli, pairs and null sets are seeded. The cropped-window recomputation equals the full network in fp64 to 10^{-14}.

Null sizes: 2000 per null for H1 and H4; 1000 for H2, the composited test and H5; 500 for the ridge readout, the seeds and the LRN check. The smallest attainable permutation p is $1/(M+1)$.

Deviations from the approved plan are listed in the pre-registration: the planned locality hypothesis became a unit test, and the secondary decision criterion was specified before the confirmatory run.

B Exploratory results

IIA by $|\Delta\theta|$ bin (30–60, 60–90, 90–120, 120–150, 150–180°): 0.111, 0.105, 0.222, 0.303, 0.149. H5 IIA by rotation angle: 0.030 at +45° and 0.140 at 180°.

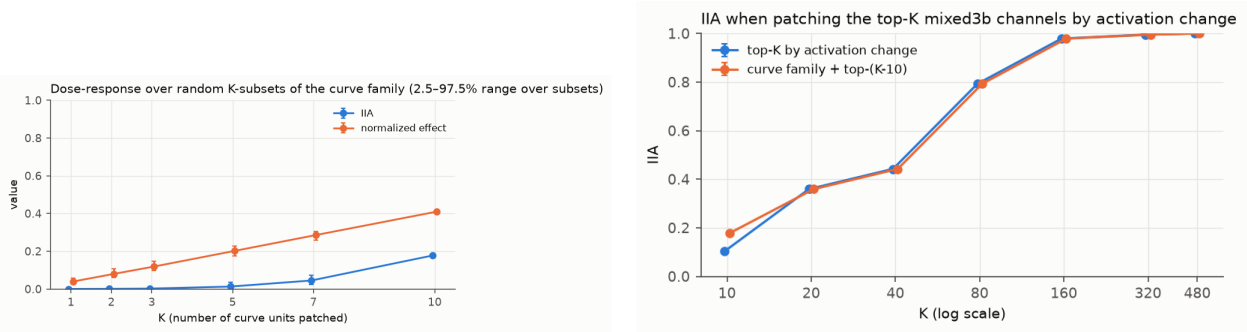


Figure 4: Left: dose-response over random K-subsets of the curve family. Right: IIA when patching the top-K `mixed3b` channels by activation change; the x axis shows $\log_2 K$.

AI statement

This codebase and paper were written by Claude Code (Anthropic's coding agent, model Claude Opus 5.5). Libor Burian provided the exact research idea that Claude Code executed: counterfactual concept-swap interventions validating a Circuits neuron explanation in a CNN. Human cleanup and rewriting are in progress.